

Automated RNA Condensate Image Analysis Pipeline

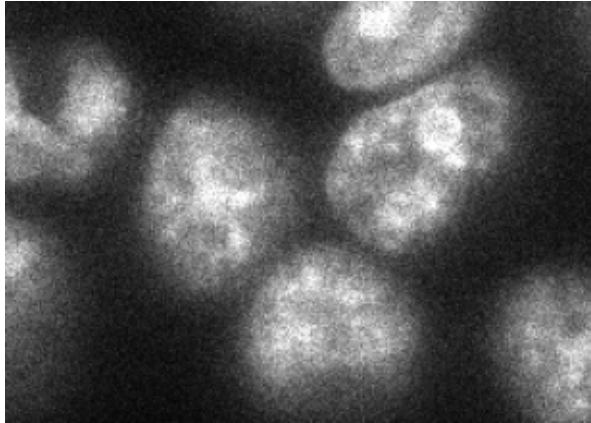
A Technical Reference

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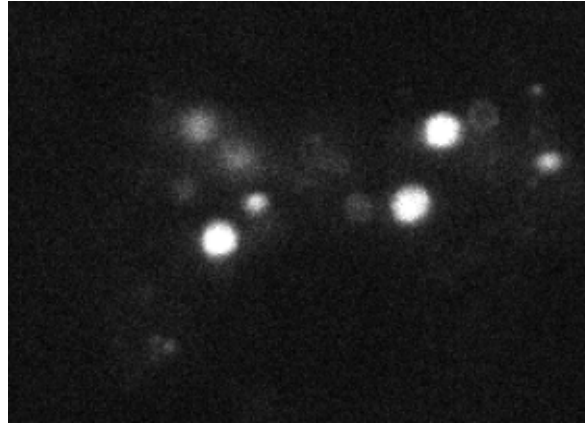
1. The Raw Images — 16-bit Z-stacks explained
2. Key Terms — voxel, instance segmentation, label mask, dilute phase, PC
3. Cellpose 3 — denoising, 3D mode, cyto3 explained
4. Mask Structure — bit ranges, label IDs, how to view in Fiji
5. Pixel Selection — how masks connect to the intensity calculation
6. Background Subtraction — why B matters and what it corrects
7. The Dilute Phase Patch — why 10x10x10 voxels
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1. The Raw Images

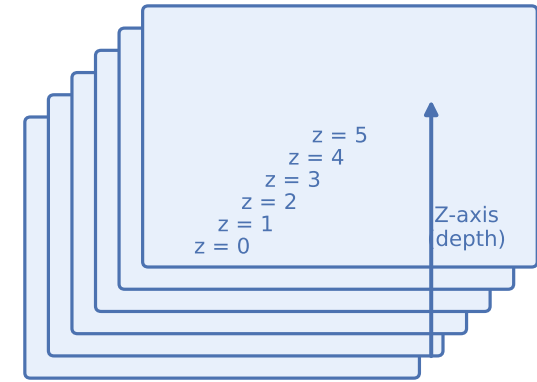
Ch1 - Nuclei (z=27)



Ch2 - Condensates (z=27)



Z-stack structure



55 slices total | 185x259 px each

Bit depth: Each pixel is a 16-bit unsigned integer — values from 0 to 65,535.

0 = no fluorescence (dark). 65,535 = maximum brightness. Condensate pixels are typically in the hundreds.

Why two channels? Ch1 (nuclei dye) tells you WHERE the nuclei are. Ch2 (Broccoli aptamer) tells you WHERE condensates are.

The PC is always computed from Ch2 intensity values. Ch1 only provides the nuclear boundary mask.

Why Z-stacks? Cells are 3D. A single 2D slice misses condensates above or below the focal plane.

55 slices at different depths capture the full cell volume, enabling true 3D segmentation and volume estimation.

2. Key Terms

Voxel

Short for volumetric pixel. A pixel is 2D (width x height). A voxel adds depth: width x height x Z-thickness.

In your dataset, one voxel = 1 XY pixel x 1 Z-slice. Counting voxels per object measures its 3D volume.

If you know the physical size (e.g. 0.065 μm XY, 0.3 μm Z), multiply to convert voxels to μm^3 .

Instance segmentation

Detecting and labelling each individual object separately. Semantic segmentation just says nucleus vs. background.

Instance segmentation says nucleus #1, nucleus #2, ... #80 — each gets its own unique integer ID.

Cellpose performs instance segmentation, which is why masks contain integer IDs, not just 0/1.

Label mask

An array with the same shape as the image (185x259 per slice). Each pixel holds an integer:

0 = background, 1 = object #1, 2 = object #2, ... up to 80 (nuclei) or 186 (condensates).

Stored as int32 (32-bit). Set Fiji range to 0-80 or 0-186 to see individual object shades.

Binary mask

A label mask collapsed to True/False: any pixel > 0 becomes True (object), 0 stays False (background).

Used in the PC calculation to select pixels: `nuc_mask = (nuclei_labels > 0)`. This is the 0-1 view in Fiji.

Dilute phase

The nucleoplasm inside the nucleus but outside any condensate — diffuse RNA/protein that has not phase-separated.

Used as the denominator in PC: represents the baseline nuclear fluorescence level.

Partition Coefficient (PC)

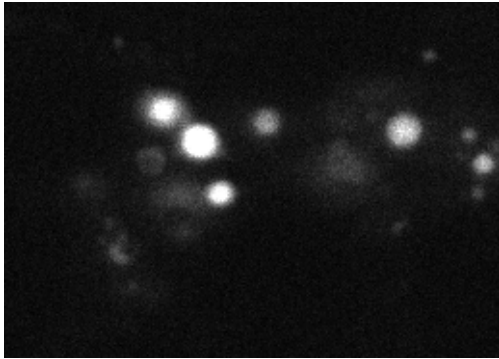
Ratio of condensate fluorescence density to dilute phase density, both background-subtracted.

PC = 1: condensates are no brighter than surroundings (no enrichment).

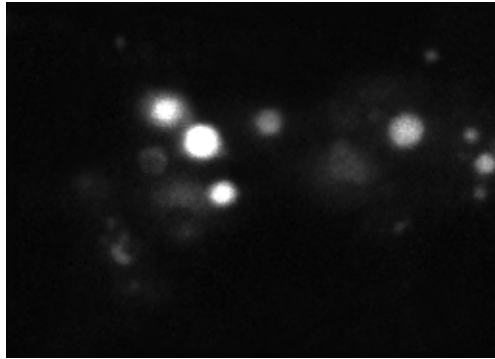
PC = 6.775: condensate voxels are $\sim 7\times$ brighter than the nuclear dilute background.

3. Cellpose 3 — How Segmentation Works

1. Raw Ch2 input (z=20)



2. After DenoiseModel



3. Cellpose 3 output (label mask)



Step 1 - DenoiseModel (denoise_cyto3)

Each Z-slice is passed through Cellpose 3's denoising neural network before segmentation.

Removes shot noise (random pixel fluctuations from the detector) while preserving biological signal.

Sharpens condensate boundaries so masks are drawn tighter around each object, improving PC accuracy.

Step 2 - 3D Segmentation (do_3D=True, cyto3)

Old pipeline (cyto2): Cellpose ran independently on each of 55 slices. Objects spanning multiple slices

could be detected inconsistently across slices (stitching artifacts). do_3D=True processes XY, XZ, YZ

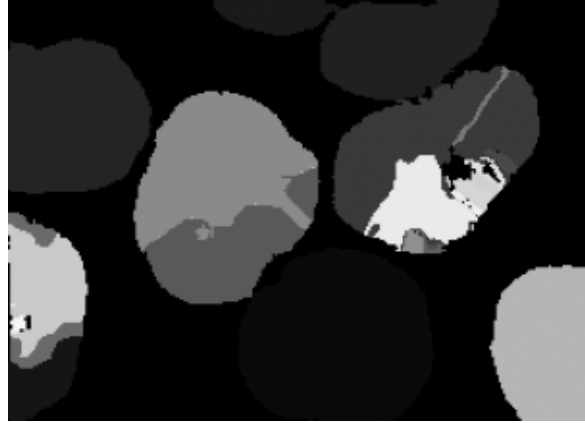
planes simultaneously, merging gradient flows across axes. Objects are coherent 3D volumes.

4. Mask Structure & Bit Ranges

Range 0-1 (binary)
white = nucleus, black = background



Range 0-48 (labeled)
each nucleus = different gray shade



Glasbey LUT
each nucleus = distinct color



What is stored in the mask file?

An int32 array (32-bit signed integer) with the same shape as the image (55 x 185 x 259).

Each pixel holds the integer ID of the object it belongs to. Background = 0. Object IDs start at 1.

Max value is 80 (nuclei) or 186 (condensates) — the total number of detected objects.

Why does Fiji show it black without adjusting the range?

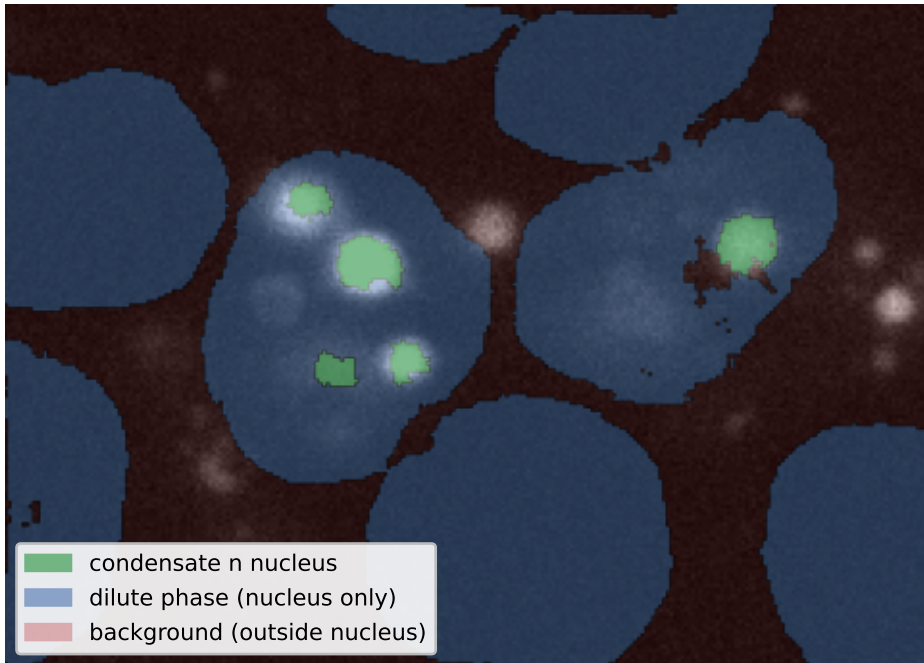
Fiji defaults to the full int32 range (-2,147,483,648 to +2,147,483,648) for 32-bit images.

Your max value of 80 or 186 is essentially 0 on that scale — everything appears black.

Fix: Image > Adjust > Brightness/Contrast > Set min=0, max=80 or 186. Or just click Auto.

5. Pixel Selection — How Masks Connect to the Calculation

Three regions at z=20



Step 1 - Convert label masks to boolean arrays

`nuc_mask = (nuclei_labels > 0)` True wherever any nucleus exists

`cond_mask = (cond_labels > 0)` True wherever any condensate exists

Both arrays are shape 185x259. Each element is True or False.

Step 2 - Define the three regions

`condensate n nucleus = cond_mask AND nuc_mask`

`dilute phase = nuc_mask AND NOT cond_mask`

`background = NOT nuc_mask`

Step 3 - Index into the raw Ch2 image

`cond_pixels = Ch2_image[condensate n nucleus]`

`dil_pixels = Ch2_image[dilute phase]`

These are lists of raw 16-bit intensity values from the original image.

6. Background Subtraction — Why It Matters

What is B?

B = minimum intensity across the entire Ch2 Z-stack = 76

This is the camera dark current + autofluorescence — signal present even with no condensate.

Every pixel has this baseline offset on top of the true fluorescence signal.

What does `clip(pixel - B, 0)` mean?

`pixel - B` subtracts the background floor from every intensity value.

`clip(..., 0)` sets any negative result to 0 (in case a pixel is slightly below B due to noise).

Result: background-corrected intensity representing only true fluorescence above the noise floor.

Why does this change the PC so dramatically?

Without subtraction (old pipeline): cond density ~ 563 dilute density ~ 148 PC = 3.8

With subtraction (new pipeline): cond density ~ 487 dilute density ~ 72 PC = 6.775

B = 76 adds equally to both in absolute terms, but 76 is a tiny fraction of 487 (condensate, bright)

yet a LARGE fraction of 72 (dilute, dim) — inflating the denominator and pulling PC toward 1.

7. The Dilute Phase Patch— Why 10x10x10 Voxels

What is a voxel (reminder)?

A voxel is one 3D unit: 1 pixel wide x 1 pixel tall x 1 Z-slice deep.

A 10x10x10 patch is a small cube of 1,000 voxels — a tiny homogeneous region of the nucleus.

Why not average ALL dilute phase pixels?

The dilute phase includes pixels near condensate edges. These edge pixels pick up bleed-through from the bright condensate next door due to the microscope point spread function (PSF).

Averaging all dilute pixels would include these contaminated edges, inflating the dilute density and deflating the PC. A patch sampled well inside the dilute phase avoids edge contamination.

How is the patch found?

1. Find all voxels that are inside a nucleus AND outside any condensate (`dilute_mask = True`).
2. For each candidate voxel, check if the full 10x10x10 cube starting there is ENTIRELY dilute.
3. Use the first valid cube found (random seed = 42 ensures reproducibility across runs).
4. Fallback: if no valid patch exists, use the mean of all dilute voxels instead.

8. Final Formula & Results

```
B = min( Ch2_stack )

rho_cond = sum( clip(pixel - B, 0) ) / N_cond_voxels

rho_dil = mean( clip(pixel - B, 0) ) over 10x10x10 patch

PC = rho_cond / rho_dil
```

Results from the spring pipeline run:

B (background floor) = 76 (min pixel across all 55 slices of Ch2)

rho_cond (condensate density) = 487.34 (mean background-subtracted intensity over nuclear condensate voxels)

rho_dil (dilute density) = 71.94 (mean background-subtracted intensity in a 10x10x10 dilute patch)

PC = 487.34 / 71.94 = 6.775

Reference PC (manual ImageJ benchmark) = 6.32

The automated pipeline recovers the manually measured benchmark value.

186 condensates and 80 nuclei detected across 55 Z-slices.

9. Full Pipeline Flowchart

